Adding External Libraries to a Java Project

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What is an external library?

Programmers often run into the same problems that other programmers have run into, which leads to many programmers re-solving the same problems that have already been solved. The goal is to not continually “reinvent the wheel”. With a library, a programmer can release code separately to allow any other application to use it. These reusable libraries save time and resources, and a goal of library creation is to make them as easy to use as possible, which often makes it quick and simple to implement them into an application.

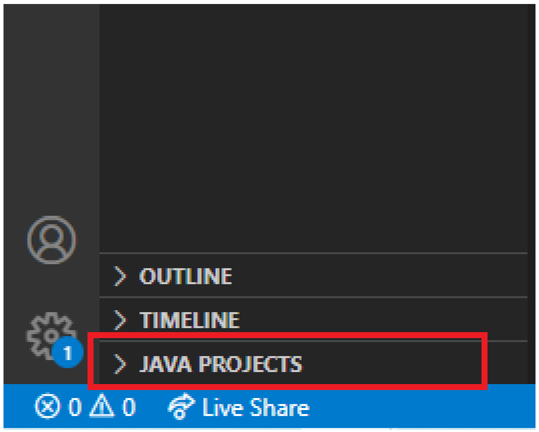
Java has many different ways to handle external library integration, but the way this guide will go over is the “manual” way of adding the library to a project’s build path. Keep in mind that this will need to be done for EVERY project that wants to use a particular external library.

How to add an external library to a Java project

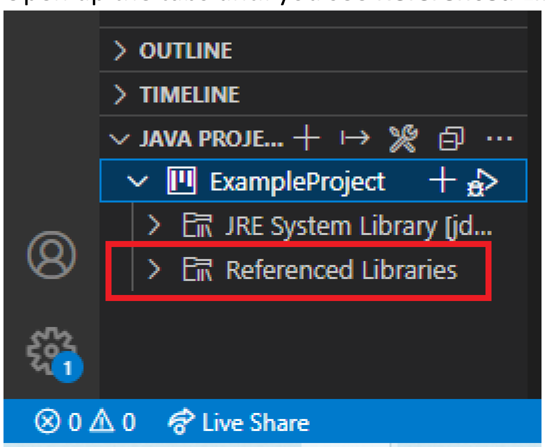
There are various ways of adding an external library to a Java project based off of the IDE you are using for your project. You can of course do it manually but…trust me and don’t do this. You can also setup a build tool like Maven, but this document will focus on adding an externally library (packaged as a **.jar** file) manually to a project. The below sections cover how to do this using Visual Studio Code, Eclipse IDE, and IntelliJ IDE.

Visual Studio Code

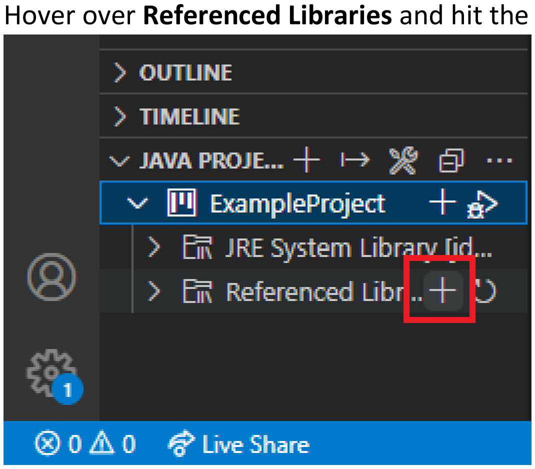
1. Open the desired Java project. If you do not have a project yet, open a desired folder and create a **.java** file in it so that Visual Studio Code will recognize it as a Java project.
2. At the bottom left of the window, you should see a tab that says **Java Projects**.



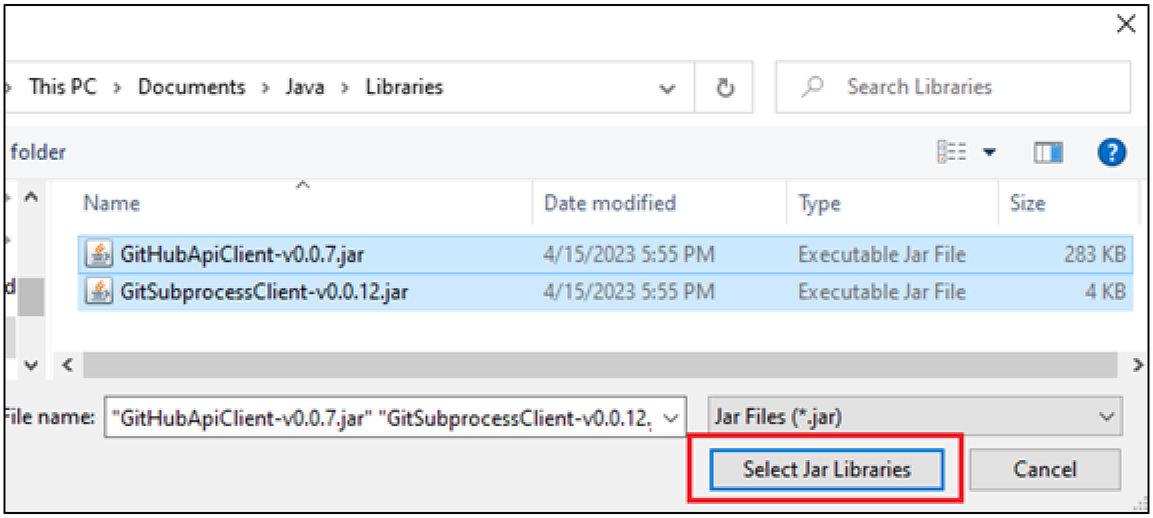
3. Open up the tabs until you see **Referenced Libraries**.



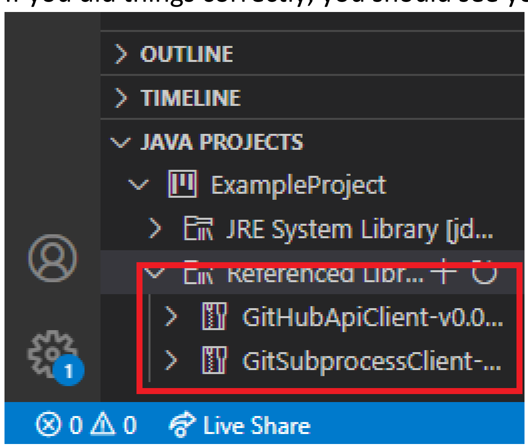
4. Hover over **Referenced Libraries** and hit the plus button **+** that shows up.



5. Upon clicking the plus button, a file dialog window will pop up. Select your desired **.jar** files and submit.



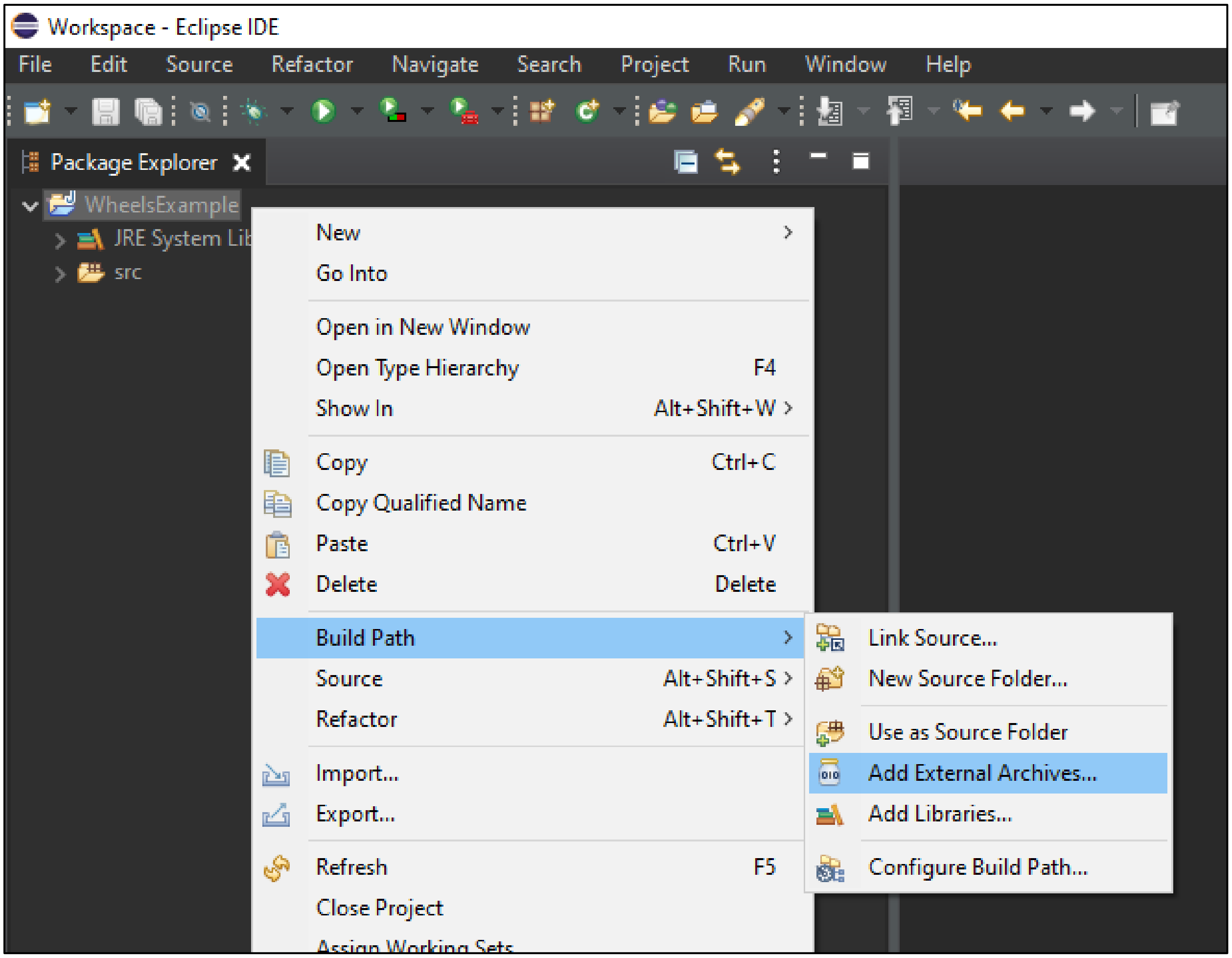
6. If you did things correctly, you should see your **.jar** files under the **Referenced Libraries** tab.



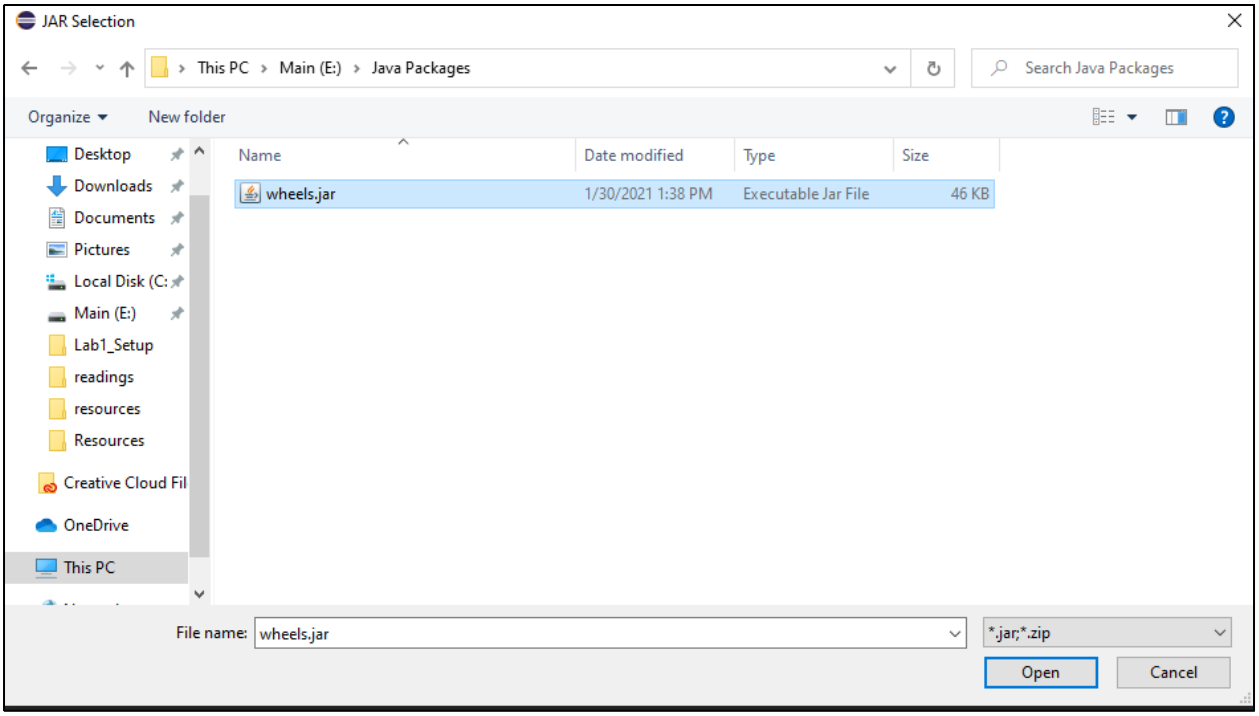
7. You can now use the library in your project. Be sure to look at documentation to learn how to use it – you will definitely have to use an import statement of some kind in your code files to use an external library.
8. Also, after importing libraries, Visual Studio Code will create a folder in the project named **.vscode** with a **settings.json** file inside of there. This file contains the file paths linking your project to the **.jar** files on your computer. Note that this file is computer specific, so if you are using version control, you should ignore this file.

Eclipse

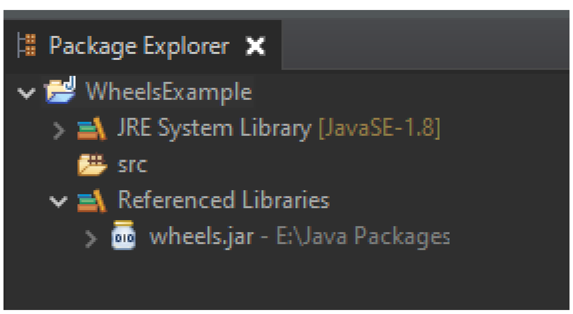
1. In your Java project, right click on the project that you want to add the external library to, select **Build Path** and then **Add External Archives…**
    a. In the below image, the project chosen is called **WheelsExample**



2. A file dialog window will pop up. Select the jar file for the library that you want to add to the project.



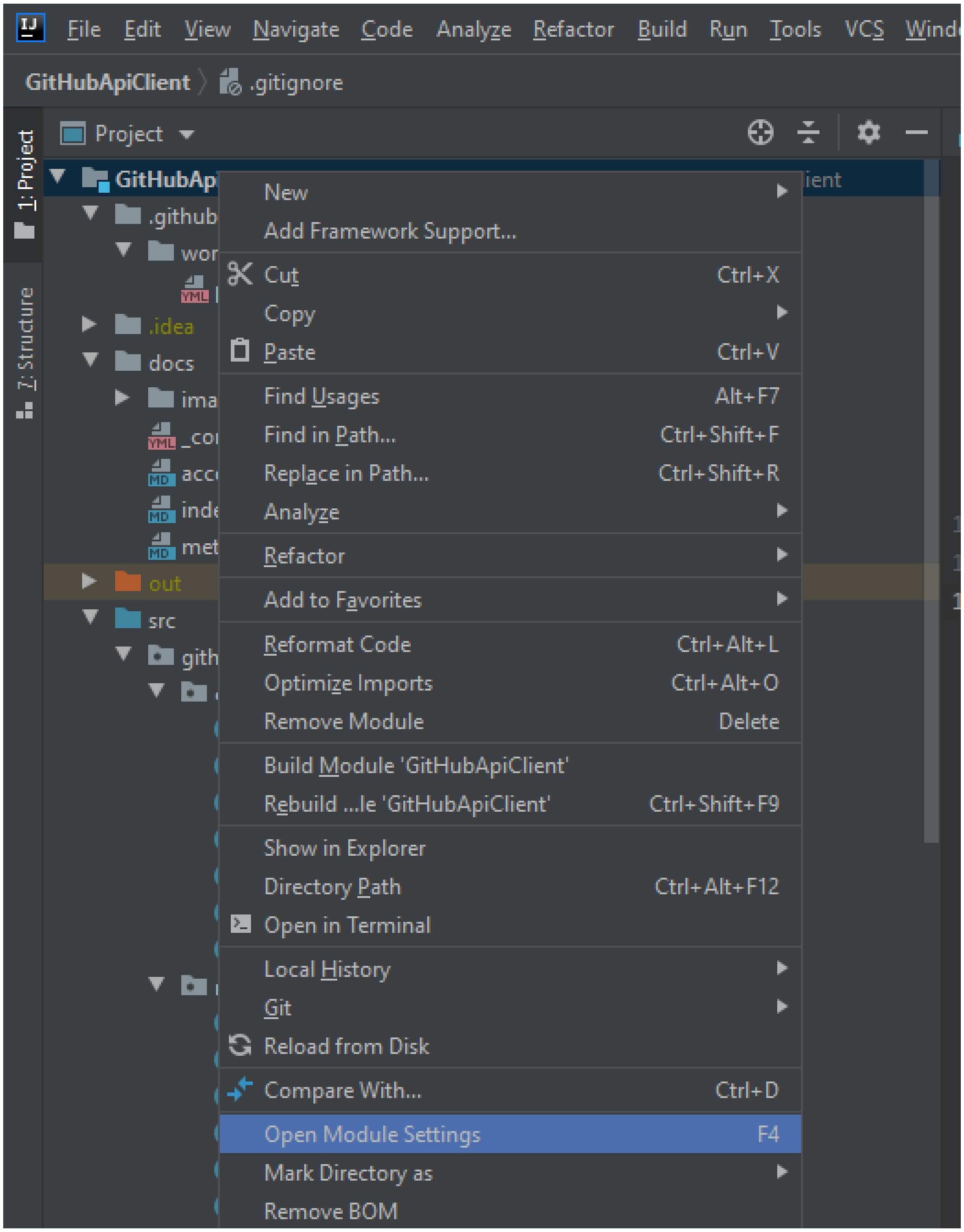
3. If you did things correctly, you should see your libraries added under the **Referenced Libraries** tab under your Eclipse project.



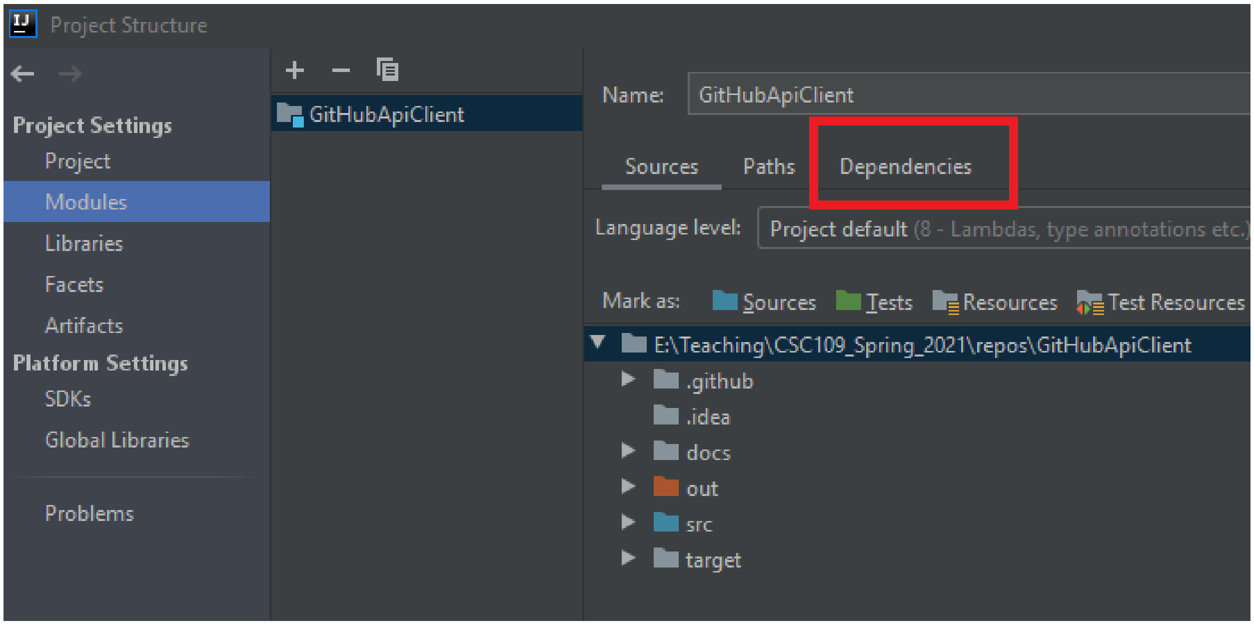
4. You can now use the library in your project. Be sure to look at documentation to learn how to use it – you will definitely have to use an import statement of some kind in your code files to use an external library.

IntelliJ

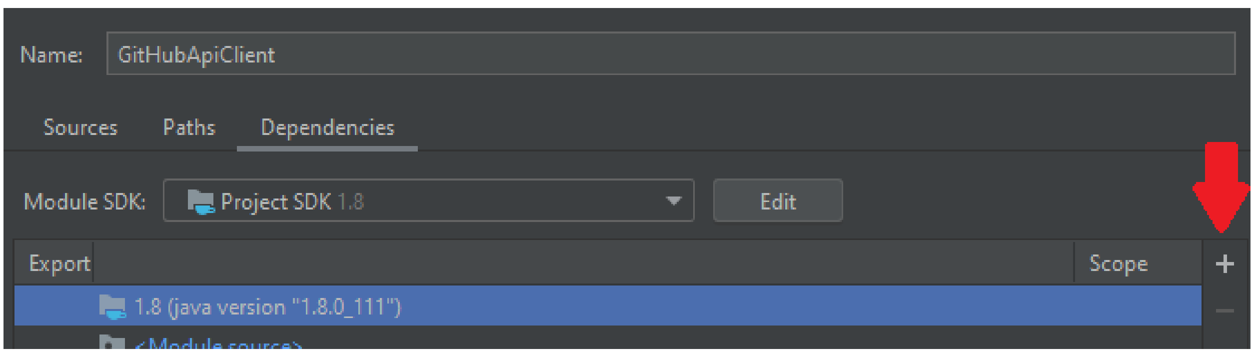
1. Right click on the root project folder and select **Open Module Settings**:



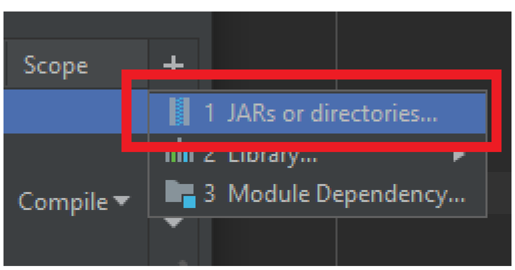
2. Click **Dependencies**:



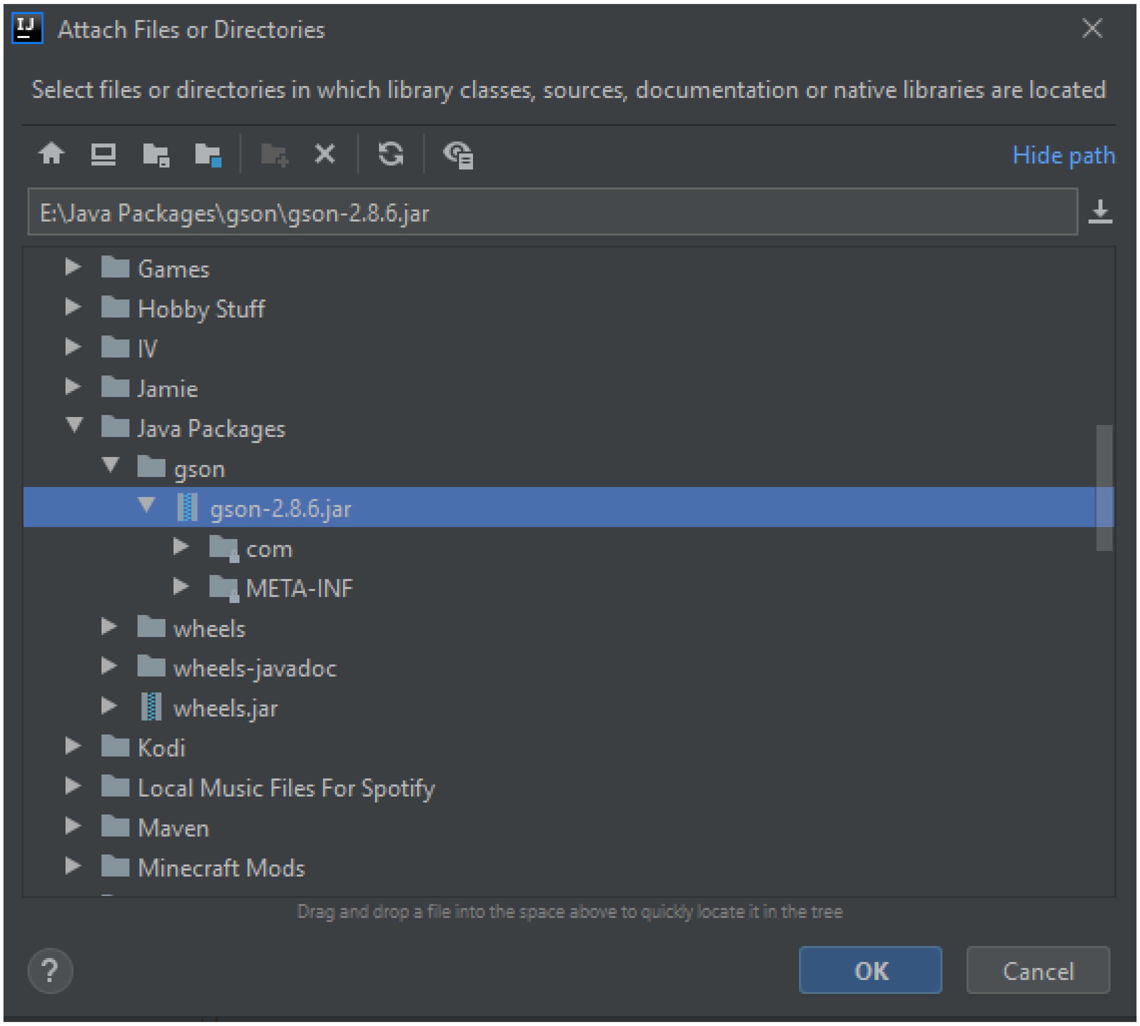
3. Press the tiny little plus button to add a dependency:



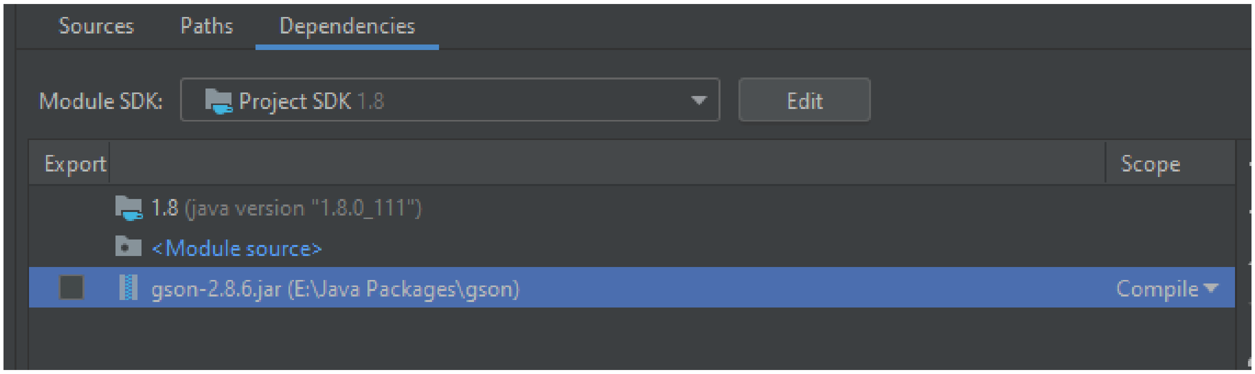
4. A menu will pop out – select **JARs or directories**:



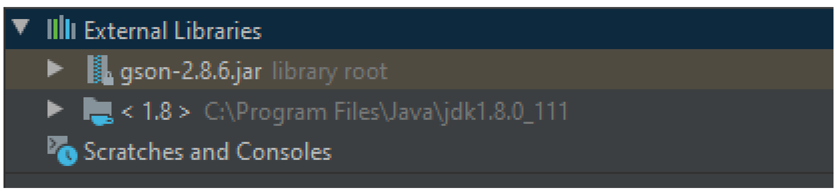
5. A file dialog window will pop up. Select the jar file for the library that you want to add to the project.



6. If you did things correctly, you should see your library in the **Dependencies** list and can close out of this window:



You'll also be able to see it in the **External Libraries** tab in the project explorer:



7. You can now use the library in your project. Be sure to look at documentation to learn how to use it – you will definitely have to use an import statement of some kind in your code files to use an external library.